Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)

# MANIPAL ACADEMY OF HIGHER EDUCATION

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SEVENTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOVEMBER 2024

**Machine Learning [ECE 4052]**

**Marks: 50** **Duration: 180 mins.**

**Descriptive**

**Answer all the questions.**

Instructions to Candidates: Answer ALL questions. Missing data may be suitably assumed.

1) Describe the Gram-Schmidt orthogonalization for multiple regression using univariate regression. Clearly give all the steps, starting from the simple univariate regression. (4)

A)

B) Consider a 0-1 loss function for a 4-class problem. Assume $Pr(g_1/x) = 0.3$, $Pr(g_2/x) = 0.4$, and $Pr(g_3/x) = 0.2$, $Pr(g_4/x) = 0.1$ ,where g1, g2, g3 and g4 are 4-class labels. Construct the loss function and find the error scores for each class. (3)

C) In a linear regression learning, show that the model parameters are given by, $\hat{\vec{\beta}} = (X^T X)^{-1} X^T Y$. Here X denotes the input data matrix and Y is the output vector. (3)

2) Explain the procedure to find the derived input data directions using principal component regression (PCR). Give the estimate of output from the derived inputs. What are limitations of PCR? (4)

A)

B) Explain the importance of Shrinkage methods. Describe the ridge regression and obtain the expression for ridge parameters. (3)

C) Show that the likelihood ratio of two classes C1 and C2 is $\frac{P(x/C_1)}{P(x/C_2)} = e^{\beta_1 x + \beta_0}$, where $\beta_1 = \frac{(\mu_1 - \mu_2)}{\sigma^2}$ and $\beta_0 = \frac{(\mu_2^2 - \mu_1^2)}{2\sigma^2}$. Assume that both the classes are Gaussian distributed with same variance. (3)

3) Illustrate logistic regression-based classifier and show that it is a linear classifier. Obtain the log-likelihood function for the same. (4)

A)

B) What are the criteria used in optimal separating classifier? Explain and obtain the objective function for this case. (3)

C) A certain disease 'D' is seen in one person out of every million. It is given that if a person has the disease, then there is 99% chance the test result will be positive; however, if a person doesn't have the disease, then there is 0.1% chance that the test result will be positive. Calculate the probability of a new patient having the disease if the test result is positive. How does the test results help in decision making? (3)

4) Define multi-layer perceptron (MLP) structure. Explain MLP in terms of basic perceptron and obtain the expression for outputs of multi-class MLP. (4)

A)

B) How the Boolean function 'OR' can be learned using perceptron? Give its structure and geometric interpretation. (3)

C) Describe decision tree and hence illustrate the Classification tree. What is the performance measure considered in decision tree? (3)

5) Describe the backpropagation algorithm for training the multi-layer perceptron (MLP). Support your description with suitable mathematical steps. (4)

A)

B) With the help of structures, describe how linear regression can be visualized using graphical models. (3)

C) Explain the Bayesian networks with suitable examples. How do you obtain the Naïve Bayes' classifier? (3)

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